

EE3621: DIGITAL SIGNAL PROCESSING | III B.TECH EEE

§ Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The **Overlap-Save (OLS)** method (also called Overlap-Discard) is an alternative block-filtering method where input blocks overlap by $M - 1$ samples. Rather than adding output tails as in OLA, the corrupted aliased samples are simply discarded from the circular convolution output.

Pedagogical Strategy:

1. Formulate input block partitioning with $M - 1$ sample overlap: $x_m[n] = x[mL + n - (M - 1)]$.
2. Explain why the first $M - 1$ points of circular convolution represent wrap-around aliasing.
3. Demonstrate that the remaining L points exactly equal the linear convolution.
4. Compare Overlap-Add and Overlap-Save:
 - OLA: Disjoint inputs $\rightarrow$ Overlapping output additions.
 - OLS: Overlapping inputs $\rightarrow$ Discard aliased output prefix; direct concatenation.
5. Explain why OLS is preferred in SIMD/GPU architectures because it requires zero output additions (pure memory copy).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** overlapping input blocks for the Overlap-Save algorithm.
2. **Execute** numerical filtering via Overlap-Save and identify aliased regions.
3. **Compare** OLA and OLS in terms of memory access, register operations, and arithmetic complexity.
4. **Select** optimal block sizes for real-time DSP implementation.

2. MATHEMATICAL FOUNDATIONS

§ 2.1 Overlap-Save Block Formulation

Let FIR filter $h[n]$ have length M . Choose block length L such that total FFT length is $N = L + M - 1$. Construct input blocks $x_m[n]$ of length N by prepending the last $M - 1$ points from block $m - 1$:

$$x_m[n] = x[mL + n - (M - 1)], \quad 0 \leq n \leq N - 1$$

For the first block ($m = 0$), prepend $M - 1$ zeros:

$$x_0[n] = \underbrace{\{0, 0, \dots, 0\}}_{M-1 \text{ zeros}}, x[0], x[1], \dots, x[L - 1]$$

§ 2.2 Circular Convolution & Discard Mechanism

Zero-pad $h[n]$ to length N . Compute the N -point circular convolution:

$$\tilde{y}_m[n] = x_m[n] \circledast_N h[n] = \text{IDFT}_N\{\text{DFT}_N\{x_m\} \cdot \text{DFT}_N\{h\}\}$$

The output contains:

- Samples $n = 0, 1, \dots, M - 2$: **Corrupted by circular wrap-around aliasing** $\implies$ **DISCARD**.
- Samples $n = M - 1, M, \dots, N - 1$: **Valid linear convolution samples** $\implies$ **SAVE**.

§ 2.3 Synthesis of Output

The total linear convolution $y[n]$ is formed by direct concatenation of the saved portions:

$$y[mL + r] = \tilde{y}_m[r + M - 1], \quad 0 \leq r \leq L - 1$$

3. WORKED NUMERICAL EXAMPLES

§ Example 14.1: Complete Overlap-Save Filtering

Problem: Filter the input sequence $x[n] = \{1, 2, -1, 2, 3, -2, 0, 1, 2, 1\}$ of length $L_x = 10$ with the FIR impulse response $h[n] = \{1, 2, 1\}$ of length $M = 3$ using the Overlap-Save method with circular FFT length $N = 6$ ($L = 4$).

Solution:

Step 1: Filter and Block Parameter Formulation

- Filter length: $M = 3$
- Number of overlap samples: $M - 1 = 2$
- Circular convolution / FFT block length: $N = 6$
- New input samples processed per block: $L = N - M + 1 = 6 - 3 + 1 = 4$
- Zero-pad the filter $h[n]$ to length $N = 6$:

$$h[n] = \{1, 2, 1, 0, 0, 0\}$$

Step 2: Construct Input Blocks of Length $N = 6$

Each block is formed by taking $M - 1 = 2$ overlap samples from the preceding block followed by $L = 4$ new input samples:

$$x_m[n] = x[mL + n - (M - 1)], \quad 0 \leq n \leq N - 1$$

- **Block 0** ($m = 0$): Prepend $M - 1 = 2$ zeros:

$$x_0[n] = \{0, 0, x[0], x[1], x[2], x[3]\} = \{0, 0, 1, 2, -1, 2\}$$

- **Block 1** ($m = 1$): Overlap the last 2 samples of block 0 ($x[2] = -1, x[3] = 2$):

$$x_1[n] = \{x[2], x[3], x[4], x[5], x[6], x[7]\} = \{-1, 2, 3, -2, 0, 1\}$$

- **Block 2** ($m = 2$): Overlap the last 2 samples of block 1 ($x[6] = 0, x[7] = 1$), zero-pad at end:

$$x_2[n] = \{x[6], x[7], x[8], x[9], 0, 0\} = \{0, 1, 2, 1, 0, 0\}$$

Step 3: Circular Convolution Formula & Mathematical Formulation

The circular convolution of two N -point sequences $x_m[n]$ and $h[n]$ is defined by:

$$\tilde{y}_m[n] = x_m[n] \otimes_N h[n] = \sum_{k=0}^{N-1} h[k] x_m[((n-k))_N], \quad 0 \leq n \leq N-1$$

where the double-parentheses index $((n-k))_N = (n-k) \bmod N$ represents the modulo- N periodic circular time-shift.

Since $h[n] = \{1, 2, 1, 0, 0, 0\}$ with $N = 6$, only $h[0] = 1$, $h[1] = 2$, and $h[2] = 1$ are non-zero. Substituting these into the formula yields the 3-tap circular difference equation:

$$\tilde{y}_m[n] = 1 \cdot x_m[((n))_6] + 2 \cdot x_m[((n-1))_6] + 1 \cdot x_m[((n-2))_6], \quad 0 \leq n \leq 5$$

In matrix form, this circular convolution corresponds to multiplying the input vector by the 6×6 circulant matrix H_c :

$$\begin{bmatrix} \tilde{y}_m[0] \\ \tilde{y}_m[1] \\ \tilde{y}_m[2] \\ \tilde{y}_m[3] \\ \tilde{y}_m[4] \\ \tilde{y}_m[5] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 2 \\ 2 & 1 & 0 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_m[0] \\ x_m[1] \\ x_m[2] \\ x_m[3] \\ x_m[4] \\ x_m[5] \end{bmatrix}$$

Notice that for $n = 0$ and $n = 1$, terms wrap around from the end of the block ($x_m[4]$ and $x_m[5]$), causing **time-domain circular aliasing**.

For $n = 2, 3, 4, 5$, no wrap-around occurs, so the output matches the exact **linear convolution**.

Step 4: Detailed Step-by-Step Block Computations

1. Block 0 ($m = 0$): $x_0[n] = \{0, 0, 1, 2, -1, 2\}$

- $n = 0$: $\tilde{y}_0[0] = x_0[0] + 2x_0[5] + x_0[4] = 0 + 2(2) + (-1) = 3 \implies$ **DISCARD (Aliased)**
- $n = 1$: $\tilde{y}_0[1] = x_0[1] + 2x_0[0] + x_0[5] = 0 + 2(0) + 2 = 2 \implies$ **DISCARD (Aliased)**
- $n = 2$: $\tilde{y}_0[2] = x_0[2] + 2x_0[1] + x_0[0] = 1 + 2(0) + 0 = 1 \implies$ **SAVE** ($y[0] = 1$)
- $n = 3$: $\tilde{y}_0[3] = x_0[3] + 2x_0[2] + x_0[1] = 2 + 2(1) + 0 = 4 \implies$ **SAVE** ($y[1] = 4$)
- $n = 4$: $\tilde{y}_0[4] = x_0[4] + 2x_0[3] + x_0[2] = -1 + 2(2) + 1 = 4 \implies$ **SAVE** ($y[2] = 4$)
- $n = 5$: $\tilde{y}_0[5] = x_0[5] + 2x_0[4] + x_0[3] = 2 + 2(-1) + 2 = 2 \implies$ **SAVE** ($y[3] = 2$)

Output for Block 0:

$$\tilde{y}_0[n] = \left\{ \underbrace{3, 2}_{\text{Discard (Aliased)}}, \underbrace{1, 4, 4, 2}_{\text{Save (Linear Conv)}} \right\}$$

2. Block 1 ($m = 1$): $x_1[n] = \{-1, 2, 3, -2, 0, 1\}$

- $n = 0$: $\tilde{y}_1[0] = x_1[0] + 2x_1[5] + x_1[4] = -1 + 2(1) + 0 = 1 \implies$ **DISCARD (Aliased)**
- $n = 1$: $\tilde{y}_1[1] = x_1[1] + 2x_1[0] + x_1[5] = 2 + 2(-1) + 1 = 1 \implies$ **DISCARD (Aliased)**
- $n = 2$: $\tilde{y}_1[2] = x_1[2] + 2x_1[1] + x_1[0] = 3 + 2(2) + (-1) = 6 \implies$ **SAVE** ($y[4] = 6$)
- $n = 3$: $\tilde{y}_1[3] = x_1[3] + 2x_1[2] + x_1[1] = -2 + 2(3) + 2 = 6 \implies$ **SAVE** ($y[5] = 6$)
- $n = 4$: $\tilde{y}_1[4] = x_1[4] + 2x_1[3] + x_1[2] = 0 + 2(-2) + 3 = -1 \implies$ **SAVE** ($y[6] = -1$)
- $n = 5$: $\tilde{y}_1[5] = x_1[5] + 2x_1[4] + x_1[3] = 1 + 2(0) + (-2) = -1 \implies$ **SAVE** ($y[7] = -1$)

Output for Block 1:

$$\tilde{y}_1[n] = \left\{ \underbrace{1, 1}_{\text{Discard (Aliased)}}, \underbrace{6, 6, -1, -1}_{\text{Save (Linear Conv)}} \right\}$$

3. **Block 2** ($m = 2$): $x_2[n] = \{0, 1, 2, 1, 0, 0\}$

- $n = 0$: $\tilde{y}_2[0] = x_2[0] + 2x_2[5] + x_2[4] = 0 + 2(0) + 0 = 0 \implies \text{DISCARD (Aliased)}$
- $n = 1$: $\tilde{y}_2[1] = x_2[1] + 2x_2[0] + x_2[5] = 1 + 2(0) + 0 = 1 \implies \text{DISCARD (Aliased)}$
- $n = 2$: $\tilde{y}_2[2] = x_2[2] + 2x_2[1] + x_2[0] = 2 + 2(1) + 0 = 4 \implies \text{SAVE } (y[8] = 4)$
- $n = 3$: $\tilde{y}_2[3] = x_2[3] + 2x_2[2] + x_2[1] = 1 + 2(2) + 1 = 6 \implies \text{SAVE } (y[9] = 6)$
- $n = 4$: $\tilde{y}_2[4] = x_2[4] + 2x_2[3] + x_2[2] = 0 + 2(1) + 2 = 4 \implies \text{SAVE } (y[10] = 4)$
- $n = 5$: $\tilde{y}_2[5] = x_2[5] + 2x_2[4] + x_2[3] = 0 + 2(0) + 1 = 1 \implies \text{SAVE } (y[11] = 1)$

Output for Block 2:

$$\tilde{y}_2[n] = \left\{ \underbrace{0, 1}_{\text{Discard (Aliased)}}, \underbrace{4, 6, 4, 1}_{\text{Save (Linear Conv)}} \right\}$$

Step 5: Output Assembly by Direct Concatenation

In Overlap-Save, the total linear convolution output requires **zero arithmetic additions**; the saved segments are concatenated directly:

$$y[n] = \{1, 4, 4, 2, \quad 6, 6, -1, -1, \quad 4, 6, 4, 1\}$$

$$y[n] = \{ \underset{\uparrow}{1}, 4, 4, 2, 6, 6, -1, -1, 4, 6, 4, 1 \}$$

Step 6: Analytical Verification via Direct Linear Convolution

The length of linear convolution is $L_{\text{total}} = L_x + M - 1 = 10 + 3 - 1 = 12$:

$$y_{\text{lin}}[n] = x[n] * h[n] = \sum_{k=0}^2 h[k] x[n - k]$$

- $y[0] = 1(1) = 1$
- $y[1] = 2(1) + 1(2) = 4$
- $y[2] = -1(1) + 2(2) + 1(1) = 4$
- $y[3] = 2(1) - 1(2) + 2(1) = 2$
- $y[4] = 3(1) + 2(2) - 1(1) = 6$
- $y[5] = -2(1) + 3(2) + 2(1) = 6$
- $y[6] = 0(1) - 2(2) + 3(1) = -1$
- $y[7] = 1(1) + 0(2) - 2(1) = -1$
- $y[8] = 2(1) + 1(2) + 0(1) = 4$
- $y[9] = 1(1) + 2(2) + 1(1) = 6$
- $y[10] = 0(1) + 1(2) + 2(1) = 4$
- $y[11] = 0(1) + 0(2) + 1(1) = 1$

$$y_{\text{lin}}[n] = \{ \underset{\uparrow}{1}, 4, 4, 2, 6, 6, -1, -1, 4, 6, 4, 1 \}$$

The Overlap-Save reconstruction matches the direct linear convolution exactly across all 12 points.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

§ Question 1 (15 Marks)

(a) Compare the Overlap-Add and Overlap-Save methods in detail. Under what conditions is Overlap-Save preferred? (7 Marks) (b) Filter $x[n] = \{2, -1, 3, 1, 2, 0, 1, 4\}$ with $h[n] = \{ \underset{\uparrow}{1}, -1 \}$ using the Overlap-Save method with $N = 4$ ($L = 3$). (8 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Detailed structural comparison covering input buffering, convolution, and output synthesis (4 Marks)
 - Explanation: OLS avoids output additions, making it ideal for DMA streaming and parallel SIMD/GPU memory copy pipelines (3 Marks)
- **Part (b):**
 - Parameter formulation: $M = 2 \implies M - 1 = 1$ overlap sample, $N = 4, L = 3$.
 - Block partitioning:
 - $x_0[n] = \{0, 2, -1, 3\}$ (prepend 1 zero)
 - $x_1[n] = \{3, 1, 2, 0\}$ (overlap $x[2] = 3$)
 - $x_2[n] = \{0, 1, 4, 0\}$ (overlap $x[5] = 0$, pad 1 zero) (2 Marks)
 - Circular convolution with $h[n] = \{1, -1, 0, 0\}$ ($N = 4$):
 - $\tilde{y}_0[n] = \{0, 2, -1, 3\} \otimes_4 \{1, -1, 0, 0\} = \{-3, \mathbf{2}, -\mathbf{3}, \mathbf{4}\}$ (Discard index 0)
 - $\tilde{y}_1[n] = \{3, 1, 2, 0\} \otimes_4 \{1, -1, 0, 0\} = \{3, -\mathbf{2}, \mathbf{1}, -\mathbf{2}\}$ (Discard index 0)
 - $\tilde{y}_2[n] = \{0, 1, 4, 0\} \otimes_4 \{1, -1, 0, 0\} = \{0, \mathbf{1}, \mathbf{3}, -\mathbf{4}\}$ (Discard index 0) (4 Marks)
 - Concatenation of saved parts (zero output additions):

$$y[n] = \{2, -3, 4, -2, 1, -2, 1, 3, -4\}$$

$$y[n] = \{ \underset{\uparrow}{2}, -3, 4, -2, 1, -2, 1, 3, -4 \}$$

(2 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

x = np.array([2, -1, 3, 1, 2, 0, 1, 4])
h = np.array([1, -1])
y = np.convolve(x, h)
print("Direct Linear Convolution:", y)
```